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Neuron. Author manuscript; available in PMC 2019 October 24.

Published in final edited form as:

Neuron. 2018 October 24; 100(2): 314–329. doi:10.1016/j.neuron.2018.10.018.

The AMPA receptor code of synaptic plasticity

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Abstract

Changes in the properties and post-synaptic abundance of AMPA-type glutamate receptors (AMPARs) are major mechanisms underlying various forms of synaptic plasticity, including long-term potentiation (LTP), long-term depression (LTD), and homeostatic scaling. The function and the trafficking of AMPARs to and from synapses is modulated by specific AMPAR GluA1-4 subunits, subunit specific protein interactors, auxiliary subunits, and post-translational modifications. Layers of regulation are added to AMPAR tetramers through these different interactions and modifications, increasing the computational power of synapses. Here we review the reliance of synaptic plasticity on AMPAR variants and propose “the AMPAR code” as a conceptual framework. The AMPAR code suggests that AMPAR variants will be predictive of the types and extent of synaptic plasticity which can occur and that a hierarchy exists such that certain AMPARs will be disproportionally recruited to synapses during LTP/homeostatic scaling-up, or removed during LTD/homeostatic scaling-down.

Introduction

Synapses in the central nervous system undergo bidirectional changes in synaptic strength, a process referred to as synaptic plasticity. These changes occur locally at individual synapses, during long-term potentiation (LTP) or long-term depression (LTD), collectively referred to as Hebbian plasticity, or globally during homeostatic scaling (Huganir and Nicoll, 2013, Turrigiano, 2008). It is widely believed that changes in synaptic strength through Hebbian plasticity form the cellular basis of learning and memory, while homeostatic scaling is an important mechanism for bidirectional regulation of neuronal excitability and for maintaining synaptic strength within a dynamic range. A primary mechanism in the control of synaptic strength during plasticity is an alteration in the number, composition and

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biophysical properties of AMPA-type glutamate receptors (AMPARs) in the post-synaptic membrane (Huganir and Nicoll, 2013, Malinow and Malenka, 2002).

AMPARs are glutamate-gated ion channels that mediate the majority of fast excitatory synaptic transmission in the brain. Four subunits, GluA1-4, assemble into tetramers to make up the core functional ion channel, with different combinations conferring unique cellular trafficking behaviors and biophysical characteristics (Shepherd and Huganir, 2007). Further, the trafficking, synapse anchoring and single-channel properties of AMPARs are regulated by subunit-specific protein binding partners and auxiliary subunits, and a large number of post-translational modifications (PTMs), including phosphorylation, ubiquitination, glycosylation, palmitoylation and S-nitrosylation (Huganir and Nicoll, 2013, Lu and Roche, 2012, Widagdo et al., 2017, Lussier et al., 2015). Moreover, many modifications and interactions have been shown to occur in combinations or to have feedback effects on neighboring modification sites or protein interactions, such that certain combinations of PTMs and interactions may be favored or inhibited.

Ongoing information processing in the brain requires fast excitatory transmission which relies on the immediate complement of post-synaptic AMPARs, whose channel properties are subject to regulation. The types, thresholds and range of synaptic plasticity available for any synapse is likewise determined by the nature of the AMPARs already at synapses which could be modified or those which can be added to synapses during potentiation or removed during depression. Therefore, both ongoing information processing and potential plasticity are subject to the types or modifications of AMPARs. Here we propose the model of an *AMPAR code of synaptic plasticity*. This code suggests that AMPAR subunit combinations, assembly in larger protein complexes, and PTMs may be a mode of information storage involved in synaptic computation, plasticity and learning and memory. In 2001, Jenuwein and Allis proposed the *histone code*, where the myriad PTMs of histone proteins, including phosphorylation, methylation and acetylation, form combinatorial effects that considerably enhance the informational capacity of the genetic code. Certain combinations of histone modifications would be highly predictive of regional chromatin structure and gene expression (Jenuwein and Allis, 2001, Taverna et al., 2007). Unlike histones which remain tightly associated with chromatin, AMPARs are highly dynamic, undergoing continuous endocytosis, exocytosis, lateral diffusion in the plane of the membrane, reversible trapping at synapses at the post-synaptic density (PSD), and modifications in channel properties (Opazo et al., 2012, Huganir and Nicoll, 2013, Malinow and Malenka, 2002). Thus, the number and function of AMPARs contained within synapses is dictated by a series of equilibriums and by individual channel properties. The dynamic nature of AMPARs suggests that the AMPAR code will function by shifting equilibriums in favor of synaptic accumulation or removal, and by altering ion channel properties. Each step of synapse targeting is subject to AMPAR subunit composition, auxiliary subunit interaction, and PTMs. If an AMPAR code exists, specific AMPAR compositions and PTMs should occur in a predictable manner and result in specific changes in AMPAR dynamics or synapse function. For each type of plasticity, certain AMPAR variants will be preferentially added or removed from synapses or undergo selective changes in channel properties. In this review, we will summarize what is known of how different AMPAR subunits, auxiliary subunits, PTMs, and protein binding partners impact synapse function and explore the emerging themes of the AMPAR code. While the

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histone code may not as yet have been “translated”, the concept has proven to be highly influential in the field of epigenetics (Taverna et al., 2007). We propose that a similar concept will serve as a valuable framework to assemble a large volume of literature on the role of AMPARs in synaptic plasticity.

Subunit combinations

AMPAR subunits contain highly conserved ligand binding and transmembrane domains that allow glutamate binding and channel pore assembly respectively, but are divergent in their extracellular amino-terminal domains and cytoplasmic carboxy-terminal tails which are subject to PTMs (Shepherd and Huganir, 2007) (Figure 1). Traditionally AMPARs have been divided into long and short tail groups: GluA1, GluA4 and a splice-variant of GluA2 (GluA2L) in the long-tail group and GluA2, GluA3 and a splice-variant of GluA4 (GluA4S) in the short-tail group (Shepherd and Huganir, 2007). Earlier studies suggest that long-tailed AMPARs are recruited to synapses in an activity-dependent manner, while short-tailed AMPARs undergo constitutive cycling in and out of synapses (Shi et al., 2001, Hayashi et al., 2000). GluA1-3 are expressed in the majority of neurons in the nervous system while GluA4 is primarily expressed early in life (Zhu et al., 2000), and in the mature brain GluA4 is primarily expressed in cerebellar granule neurons, in certain interneuron populations, certain circuits responsible for auditory processing, and is absent from the majority excitatory pyramidal neurons (Schwenk et al., 2014, von Gersdorff and Borst, 2002, Pelkey et al., 2015).

In hippocampal CA1 neurons the majority of AMPARs are made up from GluA1/2 and GluA2/3 subunit combinations, with a small contribution of GluA1 homomers (Wenthold et al., 1996). In this classic study, Wenthold et al., also detected a small amount of GluA1/3 heteromers (Wenthold et al., 1996), which have since been overlooked, but never discredited. Based on this study it has been assumed that the subunit combinations in the rest of the brain are largely similar, however, AMPAR subunit combinations have not been systematically analyzed throughout the brain, though the relative expression levels of the AMPAR subunits does show regional variations (Schwenk et al., 2014), suggesting that AMPAR composition will vary with brain region (Lu et al., 2009). As mentioned above, it has been suggested that longtailed AMPARs are primarily targeted to synapses in response to neuronal activity such as during the induction of LTP, while short tailed receptors are constitutively targeted to synapses (Shi et al., 2001, Hayashi et al., 2000). Some of these earlier studies relied on overexpression of GluA1 or GluA2 in young organotypic hippocampal slices, which results in the biased synthesis of homomeric receptors. Since the majority of AMPARs in the brain are heteromers, broad conclusions regarding long and short tailed receptors may be oversimplified.

A major distinction in AMPARs is between GluA2-containing and GluA2-lacking receptors. The vast majority of GluA2 mRNA is subjected to RNA editing converting a conserved glutamine 607 in the channel pore to arginine. The result of this modification is that GluA2 containing receptors have a linear current-voltage (I-V) relationship and are impermeable to Ca^{2+}, while GluA2-lacking receptors show high conductance and rapid decay kinetics, inward-rectification due to voltage-dependent block by intracellular polyamines and are

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permeable to Ca^{2+}, thus GluA2-lacking receptors are usually referred to as calcium-permeable AMPARs (CP-AMPARs) (Burnashev et al., 1992). CP-AMPARs, the majority of which are likely to be GluA1 homomers, are broadly detected in synapses in early development. After postnatal day 14 in rodents, CP-AMPARs are rapidly lost from synapses in the majority of neurons, and are almost undetectable in the majority of synapses onto excitatory pyramidal neurons in the mature brain. CP-AMPARs continue to be expressed at synapses in certain GABAergic neurons including parvalbumin and somatostatin-positive neurons and in auditory centers throughout life (Studniarczyk et al., 2013, Sanchez et al., 2010, von Gersdorff and Borst, 2002, Pelkey et al., 2015). Many studies have reported the appearance of CP-AMPARs at specific synapses during the induction of certain forms of synaptic plasticity including LTP, LTD and homeostatic scaling-up, or following certain behavioral manipulations such as fear conditioning or repeated cocaine administration and withdrawal (Clem and Huganir, 2010, Plant et al., 2006, Kim and Ziff, 2014, Soares et al., 2013, Sanderson et al., 2016, McCutcheon et al., 2011). These receptors have faster decay kinetics but large conductance, enhancing synaptic transmission, and may activate unique Ca^{2+}-sensitive signaling pathways which lead to an opening of a metaplastic state, in which synapses become primed for LTD or LTP (Clem and Huganir, 2010, Sanderson et al., 2016).

Auxiliary subunits

In addition to the core pore-forming AMPAR tetramer, AMPARs associate with several other proteins which have come to be recognized as auxiliary subunits. The best studied of these are the Transmembrane AMPAR Regulatory Proteins (TARPs), γ-2 (stargazin), γ-3, γ-4, γ-5, γ-7 and γ-8 (Tomita et al., 2003), and the cornichon like proteins CNIH2/3 (Schwenk et al., 2009). TARPs and CNIH2/3 regulate AMPAR channel properties and cellular and synaptic trafficking (Straub and Tomita, 2012, Maher et al., 2017, Jacobi and von Engelhardt, 2017). Biochemical and proteomic studies have isolated native AMPARs in complex with TARPs and CNIH, and further demonstrate the presence of tripartite AMPAR/TARP/CNIH complexes. AMPAR tetramers may assemble with 1-4 TARP or CNIH subunits, although the stoichiometry of native receptors may vary. CNIH2/3 seems to compete with TARPs for the same binding sites, showing that CNIH proteins regulate TARP-AMPAR interactions (Straub and Tomita, 2012, Herring et al., 2013).

TARPs contain four transmembrane domains and a highly basic cytosolic C-terminal tail that also contains a PDZ-ligand that has been shown to interact with PSD95 and other MAGUK scaffolds (Nicoll et al., 2006). TARPs have been shown to stabilize AMPARs both on the cell surface and at synapses (Bats et al., 2007, Sheng et al., 2018), and loss of TARPs results in decreased AMPAR expression, suggesting that “TARPless” AMPARs become unstable (Fukaya et al., 2006, Chen et al., 2000). TARPs significantly slow AMPAR deactivation and desensitization kinetics (Menuz et al., 2008, Cho et al., 2007), likely through interaction of extracellular loops of the TARP molecules and the ligand binding domains of AMPARs (Cais et al., 2014). The positively charged C-terminus of TARPs interacts strongly with membrane lipids, promoting the stabilization of AMPARs on the cell surface, but this TARP-lipid interaction prevents TARP binding to PSD95. The C-terminus contains multiple phosphorylation sites for CaMKII, PKC and PKA. Phosphorylation reduces the interaction with membrane lipids and facilitates interaction with PSD95, serving

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to trap the TARP/AMPAR complex with the PSD (Opazo et al., 2012, Bats et al., 2007, Sheng et al., 2018). A recent study that used knock-in mutations demonstrated that two key CaMKII phosphorylation sites on TARP γ-8, S277 and S281, are important mediators of hippocampal LTP and learning and memory (Park et al., 2016).

CNIH2/3 proteins promote the surface targeting of AMPARs and also slow the decay kinetics, similar to TARPs (Straub and Tomita, 2012, Boudkkazi et al., 2014). CNIH and TARPs likely compete for the same AMPAR binding sites (Straub and Tomita, 2012, Herring et al., 2013), and so one possible function of CNIH is to regulate the stoichiometry of TARP interaction, tuning the channel properties of the associated AMPARs. Herring et al., 2013 have shown that loss of CNIH2/3 from hippocampus led to a selective decrease in the synaptic activity of GluA1-containing receptors. TARP γ-8 effectively prevented CNIH2/3 interaction with GluA2/3 heteromeric receptors suggesting that CNIH proteins may control the subunit composition of synaptic AMPARs (Herring et al., 2013). CNIH PTMs have not been well characterized but are likely to add complexity to their role in AMPAR regulation.

The ability of the AMPAR core to interact with auxiliary subunits in a mixed stoichiometry, the selective interaction of auxiliary subunits with certain AMPAR subunit combinations, and the role of auxiliary proteins in the effect of AMPAR PTMs substantially increase the functional complexity of the AMPAR code. The remainder of this review will focus on the core subunits of the AMPAR and synaptic plasticity. The important role of AMPAR auxiliary subunits in synapse function and plasticity is an ongoing topic of active research (Brechet et al., 2017, Sheng et al., 2018).

Post-translational modifications of AMPAR subunits

AMPARs are subject to modification of the cytosolic C-terminus by phosphorylation, palmitoylation, ubiquitination, S-nitrosylation and potentially O-GlyNAcylation (Figure 1B). Phosphorylation is the most extensively characterized type of modification and many sites have been identified. AMPARs are substrates for cAMP-dependent kinase (PKA) and cGMP-dependent kinase (PKG) (GluA1 S845; GluA4 S842), protein kinase C (PKC) (GluA1 S818/S831/T840; GluA2 S863/S880), Ca^{++}/calmodulin dependent kinase II (CaMKII) (GluA1 S567/S831), Casein kinase II (GluA1 S579; GluA2 S588), PAK3 (GluA1 S863), and Src family tyrosine kinase (GluA2 Y876) (Roche et al., 1996, Barria et al., 1997a, Barria et al., 1997b, Esteban et al., 2003, Boehm et al., 2006, Chung et al., 2000, Lee et al., 2007, Lu et al., 2010, Lussier et al., 2014, Lussier et al., 2015, Hussain et al., 2015, Hayashi and Huganir, 2004). The GluA2 Y876 and S880 phosphorylation sites are conserved in GluA3 (GluA3 Y881 and S885) and are most likely also phosphorylated (Figure 1B). In particular, the best understood phosphorylations occur on GluA1, S831 and S845, and GluA2 S880. AMPAR phosphorylation can impact single channel properties and cellular trafficking during synaptic plasticity as described in greater detail below.

All four AMPAR subunits are palmitoylated at a conserved cysteine residue in the C-terminal juxtamembrane region (GluA1 C811, GluA2 C836, GluA3 C841, GluA4 C817) and a second site within the transmembrane domain TM2 (Hayashi et al., 2005). All four AMPAR subunit are also ubiquitinated. The major sites of ubiquitination on GluA1 and

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GluA2 have been mapped to GluA1 K868 and GluA2 K870/K882 (Widagdo et al., 2015, Widagdo et al., 2017, Lussier et al., 2015). GluA1 undergoes NMDA receptor (NMDAR) and nNOS-dependent S-nitrosylation on C875 (Selvakumar et al., 2013). GluA2, but not GluA1, may be modified by O-GlcNAcylation on an unidentified cytosolic serine (Taylor et al., 2014).

Many of the AMPAR PTMs have been shown to be regulated and required during synaptic plasticity, including LTP, LTD, homeostatic scaling, neuromodulator-mediated plasticity as well as during certain animal behaviors. Arguably the strongest evidence for the role of AMPAR PTMs comes from knock-in mutant mice in which specific amino acids have been mutated to prevent or mimic certain modifications. In order to understand the AMPAR code we must understand the role of each AMPAR subunit and modification as well as the various combinatorial effects. As an example, GluA1 undergoes 11 known PTMs each of which can occur independently and reversibly (7 phosphorylations, 2 palmitoylations, 1 S-nitrosylation, 1 ubiquitination), allowing for 2^{11}, or 2,048 possible modified variants of GluA1, each of which may be expected to have subtly different properties. This number is further multiplied when considering heteromeric AMPARs. It is likely that additional modifications remain to be discovered. In some cases, it has been shown that certain AMPAR PTMs can influence others (Figure 1C), either by inhibiting or promoting the neighboring modification, suggesting that of all possible AMPARs variants some will be favored. In the following sections we will review the role of AMPAR PTMs in different forms of synaptic plasticity and explore the emerging themes of the AMPAR code.

The AMPAR code and LTP

GluA1 and LTP—The major form of LTP identified in the brain, NMDAR-dependent LTP, is the result of increased AMPAR density in the post-synaptic membrane and/or single channel conductance of the AMPAR (Huganir and Nicoll, 2013, Herring and Nicoll, 2016, Poncer et al., 2002, Benke et al., 1998). The increase in AMPAR content within the PSD is supplied by immediately available non-synaptic pools of diffusing AMPARs within the plane of the plasma membrane, which in turn are replenished by exocytosis of intracellular pools of AMPARs localized in recycling endosomes. The exocytosis, lateral diffusion, PSD trapping and changes in single channel properties are all subject to regulation by PTMs and protein interactions (Huganir and Nicoll, 2013, Anggono and Huganir, 2012, Lu and Roche, 2012).

GluA1 directly interacts with the cytoskeletal scaffold protein 4.1N through the C-terminal juxtamembrane region, and this interaction promotes activity-dependent exocytosis of GluA1 from intracellular endosomes (Lin et al., 2009, Anggono and Huganir, 2012). Palmitoylation of GluA1 C811 prevents interaction with 4.1N and reduces PKC-mediated phosphorylation of nearby S818 (Figure 1C) (Lin et al., 2009). Conversely, phosphorylation of GluA1 S818 increases interaction with 4.1N and is necessary for activity-dependent exocytosis (Lin et al., 2009). During hippocampal LTP, PKC is activated to increase GluA1 S818 phosphorylation and a peptide mimicking phospho-GluA1 S818 or shRNA-mediated knock-down of 4.1N both lead to decay of hippocampal LTP back to baseline, suggesting

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that the PKC/4.1N mediated increase in GluA1 exocytosis is necessary to maintain LTP (Lin et al., 2009, Boehm et al., 2006).

Phosphorylation of GluA1 S831 and S845 are both strongly associated with LTP (Huganir and Nicoll, 2013, Lee et al., 2003, Barria et al., 1997b, Lee et al., 2000). NMDAR-dependent LTP requires Ca^{2+} influx leading to activation of CaMKII/PKC which directly phosphorylate GluA1 S831, increasing GluA1 single channel conductance and promoting GluA1 targeting to the PSD (Mammen et al., 1997, Barria et al., 1997a, Derkach et al., 1999, Kristensen et al., 2011, Barria et al., 1997b). PKA can be activated by Ca^{2+}-sensitive adenylcyclase or downstream of neuromodulators signaling through their G_s-coupled receptors, such as the β–adrenergic receptor or D1-type dopamine receptors, leading to phosphorylation of GluA1 S845 (Joiner et al., 2010, Sun et al., 2005). Phosphorylation of S845 increases single channel open–probability and also promotes GluA1 targeting or retention to the cell surface (Banke et al., 2000, Oh et al., 2006, Man et al., 2007). Mice containing knock-in mutations in S831 and S845 (S831A/S845A) that eliminate phosphorylation of these sites lack LTD (see below) and have partially impaired hippocampal LTP (Lee et al., 2003). However, the phosphorylation of GluA1 S831 and S845 can occur completely independently and while single S845A knock-in mutation still eliminated LTD, the single S831A or S845A knock-in mutations do not impair LTP of the Schaffer collateral-CA1 synapses in the hippocampus suggesting a synergistic role of these two phosphorylation sites on LTP expression (Lee et al., 2010). Under basal conditions in cultured neurons and in mouse forebrain, approximately 15-20% of GluA1 is phosphorylated at S831 or S845, but only a small fraction of GluA1 is dually phosphorylated (Diering et al., 2016). The phosphorylated fraction of total GluA1 can increase to close to 50% upon strong stimulation of PKC/PKA. Similarly, exposure of mice to an enriched environment also greatly increases the population of GluA1 showing dual S831/S845 phosphorylation (Figure 1C) (Diering et al., 2016). Compared to total membrane fractions, phospho-S831 receptors are enriched in the PSD under basal conditions whereas phospho-S845 receptors are not (Diering et al., 2016), suggesting that S831 phosphorylation plays a role in targeting to the PSD or stabilization within the PSD. Phosphorylation of GluA1 S845 plays a role in promoting GluA1 targeting to the cell surface and/or stabilization of GluA1 on the cell-surface by enhancing GluA1 recycling and limiting endocytosis (Ehlers, 2000, Oh et al., 2006, Man et al., 2007). Neuromodulators, such as noradrenaline, can lower the threshold for LTP and this effect is blocked by GluA1 phospho-deficient S831A/S845A knock-in mutations (Hu et al., 2007), whereas GluA1 phospho-mimetic S831D/S845D knock-in mutations show a lowered threshold for LTP that occludes the effect of noradrenaline (Makino et al., 2011). These findings, together with several previous studies, support a model where PKA-mediated phosphorylation of GluA1 S845 promotes the surface trafficking of the receptor, lowering the threshold needed to induce LTP, however, additional signaling from CaMKII is needed to promote GluA1 targeting to the PSD for maximal LTP (Figure 1) (Sun et al., 2005, Oh et al., 2006, Esteban et al., 2003).

Although LTP of SC-CA1 hippocampal synapses was normal in single GluA1 S831A or S845A knock-in mice (Lee et al., 2010), synaptic potentiation at other synapses was found to be sensitive to mutations at either of these sites. Serotonin and selective-serotonin reuptake inhibitor type anti-depressants (SSRIs) increase the phosphorylation of GluA1

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S831 and induce synaptic potentiation in the hippocampal CA1 temporoammonic pathway. This potentiation, as well as the effects of SSRIs, are absent in the GluA1 S831A mice (Cai et al., 2013), while ketamine treatment, which is being developed as a rapid acting anti-depressant, requires phosphorylation of GluA1 S845 for its effect (Zhang et al., 2016). Neuromodulator-mediated spike-timing-dependent LTP in layer IV to II/III synapses in the visual cortex and the expression of network LTP in the anterior cingulate cortex (ACC) is absent in the GluA1 S845A knock-in mice (Song et al., 2017, Seol et al., 2007). Thus, while phosphorylation of GluA1 S831 or S845 are clearly involved in LTP, the strict requirement for each site as well as the synergistic effects of dual phosphorylation are likely synapse- and induction-specific.

Other PTMs of GluA1 have also been implicated in LTP but these have not been systematically studied. S-nitrosylation of GluA1 C875 by nNOS was found to enhance the phosphorylation of GluA1 S831 by CaMKII (Figure 1) (Selvakumar et al., 2013), suggesting an additional synergistic relationship between S831 phosphorylation and C875 nitrosylation. The first intracellular loop of GluA1 can be phosphorylated by casein kinase IIβ at S579 to promote surface targeting (Lussier et al., 2014), and GluA1 S863 was found to be a target for PAK3 which promotes targeting of GluA1 to the cell surface by relieving an unidentified intracellular retention interaction (Hussain et al., 2015). However, whether these GluA1 PTMs play a role in LTP remains to be determined. Neuronal activity can also induce ubiquitination of AMPARs (Widagdo et al., 2015) although the role of this AMPAR PTM in LTP is unknown. Interestingly, ubiquination and phosphorylation of GluA1 reciprocally inhibit each other indicating that a complex cross talk of GluA1 PTMs may regulate LTP expression (Guntupalli et al., 2017). Signaling from NMDARs can activate USP8, a de-ubiquitinating enzyme that acts on AMPARs (Scudder et al., 2014), potentially limiting AMPAR ubiquitination during LTP.

The role of CP-AMPARs, likely GluA1 homomers, in LTP remains controversial. At most synapses onto glutamatergic neurons, synaptic levels of CP-AMPARs are very low. However, it has been reported that CP-AMPARs are present at peri-synaptic sites and can be rapidly and transiently recruited to synapses immediately following the induction of LTP (He et al., 2009, Plant et al., 2006, Guire et al., 2008, Yang et al., 2010). Other studies have refuted this observation (Adesnik and Nicoll, 2007, Gray et al., 2007). The presence of perisynaptic CP-AMPARs as well as their recruitment to synapses requires signaling from PKA and the S845 phosphorylation site (He et al., 2009, Clem and Huganir, 2010, Sanderson et al., 2016). While the induction protocols vary somewhat between these studies, recent data offer some explanation for the discrepancy. Sanderson et al., show that in acute hippocampal slices from mice 11-14 days old, LTP of SC-CA1 synapses can be blocked by IEM1460, an inhibitor of CP-AMPARs, whereas when slices are prepared from 17-21 day old mice LTP is insensitive to IEM1460 (Sanderson et al., 2016). These findings show that the involvement of CP-AMPARs in hippocampal LTP may follow a very narrow developmental window of only a few days. Another study has found that CP-AMPARs are present at excitatory synapses on cortical layer V neurons in mice during their active phase at night, but that these CP-AMPARs are removed during the sleep phase during the day (Lante et al., 2011). As the majority of research is done during the light phase, when CP-

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AMPARs are less prevalent, this finding suggests that some of the controversy regarding CP-AMPARs may also be subject to time of day effects.

While the importance of the GluA1 C-terminal tail and its PTMs in LTP has been questioned by molecular replacement studies using C-terminal deletions of GluA1 and GluA2 (Granger et al., 2013), recent data from Zhou et al., provide strong evidence that the GluA1 C-terminal tail is both necessary and sufficient for hippocampal LTP (Zhou et al., 2017). In this study, the authors generated mouse lines in which the endogenous C-tail of GluA1 was replaced with the C-tail of GluA2: GluA1^{C2KI}, or vice versa: GluA2^{C1KI}. Basal synaptic transmission was completely unaltered in these mutant mice, unlike previously studied genetic manipulations of AMPAR subunits (Lu et al., 2009). Importantly, LTP was completely absent in GluA1^{C2KI} mice, demonstrating that the GluA1 C-tail is necessary for LTP. In support of this conclusion, GluA2^{C1KI} mice, which express extra copies of the GluA1 C-tail, showed enhanced LTP. Incredibly, LTP was rescued in GluA1^{C2KI} and GluA2^{C1KI} double mutant mice, showing that the C-tail of GluA1 is sufficient for LTP expression, even when it is attached to the GluA2 subunit (Zhou et al., 2017). The differences in the findings of the Granger et al. and the Zhou et al. studies may be due to the techniques used. In the Granger et al. study, AMPAR GluA1-3 subunits were conditionally eliminated during development and replaced with AMPA receptors lacking the C-terminal tails. This method would dramatically disrupt the composition of synapses and may not reflect the physiological mechanisms underlying LTP expression (Sheng et al., 2013). In contrast, the Zhou et al. study used a more subtle approach by generating knock-in mutations in the C-terminal tails of GluA1, and these experiments, in addition to the GluA1 phosphorylation site knock-in experiments described above, likely reveal the critical role of the GluA1 C-terminal tails in LTP.

While much of the work described above has focused on the role of GluA1 C-terminal tails, recent studies have identified an important role for the GluA1 amino terminal domain (Fig. 1A) in synaptic trafficking and plasticity. The amino terminal domain is highly divergent between AMPAR subunits with ~56% identity. Data show that the amino terminal domain is not necessary for the formation of functional ion channels, or delivery onto the cell surface, but is required for targeting and retention of GluA1 at synapses during LTP (Watson et al., 2017, Diaz-Alonso et al., 2017). Interestingly, these recent studies support earlier models describing the activity-dependent synapse recruitment of GluA1 vs. the constitutive synapse recruitment of GluA2 (Shi et al., 2001), although not via the earlier described long vs. short cytoplasmic tails.

GluA2/3 and LTP—While the majority of the literature supports a clear role for GluA1 in LTP, GluA2/3 also participate. Earlier models of AMPAR trafficking suggested that long-tailed AMPARs (GluA1) are rapidly recruited to synapses during LTP while short-tailed AMPARs (GluA2/3) are recruited to synapses in a constitutive manner and gradually replace GluA1-containing receptors at potentiated synapses (Shi et al., 2001). This subunit exchange that occurs following LTP has been suggested as an important component in memory consolidation. LTP in the mature hippocampus is mediated by GluA2-containing receptors (Adesnik and Nicoll, 2007). GluA2 and GluA3 form direct protein interactions with GRIP1/2 and PICK1 through their PDZ-domains (Anggono and Huganir, 2012), which

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facilitate interactions with additional proteins including KIBRA, Grasp1 and NEEP21, forming a larger complex involved in AMPAR trafficking (Anggono and Huganir, 2012). Mice lacking PICK1, KIBRA or Grasp1 all show reductions in hippocampal LTP as well as defects in hippocampus-dependent learning and memory tasks (Makuch et al., 2011, Volk et al., 2010, Chiu et al., 2017) and disruption of the GRIP1-NEEP21 interaction also impairs LTP (Alberi et al., 2005, Steiner et al., 2005). An emerging theme from these studies suggests that endosomal pools of GluA2 or GluA3, possibly in the form of GluA1-containing heteromeric receptors, are mobilized to the cell-surface to facilitate LTP. Another parallel mechanism suggests that PICK1 is important during the early phase of LTP where it acts to sequester GluA2-containing receptors within endosomes, enabling CP-AMPARs to be transiently incorporated into synapses (Jaafari et al., 2012, Clem et al., 2010).

GluA3 and LTP—Recently, it has been found that LTP in cerebellar Purkinje neurons as well as cerebellar-dependent adaptation of the vestibulo-ocular reflex both depend on GluA3- but not GluA1-containing AMPARs (Gutierrez-Castellanos et al., 2017). This form of cerebellar LTP does not appear to involve trafficking of GluA3, but rather relies on an increase in the open-probability of GluA3-containing receptors through a cAMP-dependent pathway (Gutierrez-Castellanos et al., 2017). This regulation of GluA3-mediated transmission has also been observed in hippocampal CA1 neurons (Renner et al., 2017). Here it was shown GluA3 containing receptors are mostly inert under basal conditions due to a very low-open probability, although these receptors are robustly targeted to synapses (Renner et al., 2017). A recent proteomics study has even found that GluA3 subunits are the most enriched within the PSD under basal conditions (Pandya et al., 2017). Therefore, synaptic potentiation through GluA3 may not require AMPAR trafficking, but rather cAMP-dependent enhancement of channel open-probability of GluA3-containing receptors already localized to synapses, allowing for very rapid changes in synaptic efficacy. Whether this mechanism contributes to forms of hippocampal LTP remains to be determined.

GluA4 and LTP—GluA4 is expressed broadly in neurons early in life and may have unique plasticity rules relevant for synapse maturation and synapse un-silencing (Zhu et al., 2000). GluA4 S842 is phosphorylated by PKA (Esteban et al., 2003). Unlike GluA1 where the activity of PKA and CaMKII are coordinated for synaptic recruitment, PKA phosphorylation of GluA4 alone is sufficient for synapse delivery and retention (Esteban et al., 2003). The period of early postnatal life when GluA4 is expressed also coincides with a period where PKA, and not CaMKII, is required for hippocampal LTP (Yasuda et al., 2003). The fast kinetics of GluA4 assist in the processing of auditory information (von Gersdorff and Borst, 2002, Sanchez et al., 2010). The presence of synaptic GluA4 in these auditory centers could have interesting implications for the types and extent of synaptic plasticity at these synapses. GluA4 clustering at excitatory synapses, particularly onto parvalbumin-positive interneurons, is mediated by the extracellular neuronal pentraxins which bind to the extracellular amino terminal domain, again highlighting the role of extracellular interactions in AMPAR synaptic recruitment (Pelkey et al., 2015, O'Brien et al., 1999, Sia et al., 2007).

To summarize, LTP requires an enhancement of synaptic AMPARs (Huganir and Nicoll, 2013). The threshold and nature of induction as well as the magnitude and stability of LTP is

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therefore the product of the types of AMPARs which are available for synaptic recruitment. The AMPAR code proposes that a hierarchy exists for the selective incorporation of AMPAR subunit/PTM combinations, such that each form of LTP will favor the recruitment of specific AMPAR types (Figure 2). Spike-timing and noradrenaline-dependent LTP in the cortex, for example, will preferentially recruit phosphorylated S845 GluA1-containing receptors (Seol et al., 2007). Hippocampal LTP in very young neurons may preferentially recruit CP-AMPARs (Plant et al., 2006, Sanderson et al., 2016). We propose that all forms of LTP will show a hierarchy of AMPAR recruitment. As described above, LTP has been shown to occur in the absence of certain AMPAR subunits (Granger et al., 2013). Our proposed hierarchy suggests that in the absence of preferred AMPAR variants, less preferred variants, or even kainate receptors, may substitute, although this is may be only the case for the most robust LTP induction protocols (Granger et al., 2013).

The AMPAR code and LTD

GluA2/3—Several major forms of LTD described in the brain including, cerebellar LTD and NMDAR-dependent and mGluR-dependent LTD, are expressed by a decrease in synaptic AMPAR content (Huganir and Nicoll, 2013). AMPARs must be removed from the PSD, allowing for lateral diffusion away from the synapse, and concomitantly, AMPARs are locally depleted from the cell-surface by clathrin- and dynamin-dependent endocytosis (Shepherd and Huganir, 2007).

GluA2 plays a prominent role in LTD. Several studies have demonstrated that GluA2 is required for the expression of cerebellar LTD. Specifically, phosphorylation of GluA2 S880 is required as knock-in mutations that prevent phosphorylation of this site in mice (GluA2 K882A) eliminates cerebellar LTD (Chung et al., 2000, Steinberg et al., 2006). This effect of S880 on LTD is likely mediated by regulating the interaction of GluA2 with the PDZ domain containing proteins, GRIP1/2 and PICK1 that regulate AMPAR membrane trafficking (see below). Consistent with this model, genetic deletion of GRIP1/2 or PICK1 eliminates cerebellar LTD (Takamiya et al., 2008, Steinberg et al., 2006). The GluA2 S880 phosphorylation site has also been shown to be important in hippocampal LTD (Seidenman et al., 2003, Chung et al., 2000, Steinberg et al., 2006). These studies support a model where GRIP1/2 bind directly to the PDZ ligand in the GluA2/3 C-terminal tail, anchoring the receptors in the PSD (Anggono and Huganir, 2012). Phosphorylation of GluA2 S880 prevents GRIP1/2 binding and promotes the interaction of the GluA2 PDZ ligand with PICK1 (Seidenman et al., 2003, Chung et al., 2000, Steinberg et al., 2006), destabilizing GluA2 within the PSD and promoting GluA2 endocytosis. Via its BAR-domain, PICK1 can either promote the invagination of the plasma membrane or direct GluA2 to regions of invagination where clathrin-mediated endocytosis occurs (Xia et al., 2000, Fiuza et al., 2017). This mechanism likely also applies to GluA3, which also interacts with GRIP1/2 and PICK and the phosphorylation site, S885 is conserved with GluA2 S880 (Figure 1B).

Several studies have also implicated tyrosine phosphorylation of GluA2 in LTD. The GluA2 and GluA3 distal C-terminus contains a cluster of closely spaced tyrosine residues (3 in GluA2 and 4 in GluA3). GluA2 Y876 has been identified as a target for Src family tyrosine kinases (Hayashi and Huganir, 2004). GluA3 is also tyrosine-phosphorylated at the

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homologous site Y881 (unpublished observations). Glutamate receptor agonists, including glutamate, AMPA and NMDA, induce GluA2 tyrosine phosphorylation which disrupts the interaction between GluA2 and GRIP1/2 but not PICK1 (Hayashi and Huganir, 2004). Multiple studies have shown both in slice physiology experiments and *in vivo* that treatment with a short peptide mimicking the tyrosine-rich region of GluA2 (called the 3Y peptide) completely blocks LTD as well as activity-dependent GluA2 endocytosis (Fox et al., 2007, Ahmadian et al., 2004), while having no effect on basal synaptic transmission. Interestingly, hippocampal mGluR-mediated LTD involves the dephosphorylation of GluA2 Y876 and is completely blocked by inhibitors of tyrosine phosphatases (Moult et al., 2006, Gladding et al., 2009). BRAG2, an Arf6 guanine nucleotide exchange factor (GEF), has been shown to bind to the GluA2 Y-rich region (Scholz et al., 2010). During hippocampal mGluR-LTD, concomitant ligand binding and dephosphorylation of GluA2 Y876 stimulates BRAG2 GEF activity, leading to activation of Arf6 driving GluA2 endocytosis (Scholz et al., 2010). BRAG2 is also required for NMDAR-LTD, but likely through a distinct mechanism as NMDAR-LTD does not appear to involve GluA2 Y876 dephosphorylation (Moult et al., 2006), which is required for BRAG2 GEF activity (Scholz et al., 2010). In cerebellar Purkinje neurons, phosphorylation of GluA2 Y876 has been shown to inhibit subsequent phosphorylation of GluA2 S880 (Figure 1) (Kohda et al., 2013). Dephosphorylation of GluA2 Y876 by tyrosine phosphatase PTPMEG was necessary for S880 phosphorylation and expression of cerebellar LTD (Kohda et al., 2013). It is possible that phosphorylation of GluA2 Y876 and S880, as well as GluA3 Y881 and S885, might be similarly coordinated during NMDAR- or mGluR-LTD in other brain areas, but this remains to be determined.

GluA2 is also a target of ubiquitination on K870 and K882, which occurs in response to neuronal activity and is dependent on AMPAR internalization (Widagdo et al., 2015). Although the target site of ubiquitination of GluA3 was not identified, GluA3 K887 is conserved with GluA2 K882. The ubiquitination sites of GluA2 (and likely GluA3) are very close the Y876/S880 phosphorylation sites (Figure 1B and C), however, whether these PTMs are coordinated has not been determined. Further, the role of AMPAR ubiquitination in LTD remains to be determined.

Zhou et al., have recently reported that knock-in mice in which the GluA2 C-tail is replaced with the GluA1 C-tail: GluA2^{C1KI}, are completely impaired in hippocampal NMDAR-dependent LTD (Zhou et al., 2017). NMDAR-LTD can be rescued in the GluA1^{C2KI}/GluA2^{C1KI} double mutants, showing that the GluA2 C-tail is both necessary and sufficient for hippocampal NMDAR-LTD. Interestingly, in this study, mGluR-dependent LTD was not affected by the GluA1/2 C-tail exchange.

GluA1—While the literature on LTD indicates a dominant role of the GluA2 subunit, GluA1 has also been shown to be important. GluA1 S845 dephosphorylation is critical for hippocampal LTD, as the GluA1 S845A mutation results in impaired LTD (Lee et al., 2003, Lee et al., 2010), indicating that the induction of LTD requires PKA phosphorylation of GluA1 even though receptors are eventually dephosphorylated (Huganir and Nicoll, 2013). In hippocampal CA1 neurons, synaptic levels of CP-AMPARs are low. However, blocking glutamate re-uptake and causing glutamate spillover reveals a population of CP-AMPARs in the peri-synaptic region, which is completely abolished by GluA1 S845A mutation (He et

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al., 2009). It has been recently shown that this population of peri-synaptic CP-AMPARs (GluA1 homomers) are transiently recruited to the synapse during the induction of LTD in a PKA-dependent manner (Sanderson et al., 2016). During the induction phase of LTD, transient signaling from synaptic CP-AMPARs is required for the full expression of LTD, and during the progression of LTD induction these CP-AMPARs signal their own removal through activation of calcineurin (CaN) (Sanderson et al., 2016). If PKA is not able to phosphorylate GluA1 S845, CP-AMPAR recruitment to synapses during LTD is blocked, possibly explaining the complete absence of LTD in GluA1 S845A knock-in mice. AKAP5 (also called AKAP79/AKAP150) is a signaling scaffold protein required for LTD (Sanderson et al., 2016, Jurado et al., 2010). AKAP5 anchors both PKA and CaN to GluA1 allowing for the efficient coordination of these signaling pathways during LTD (Sanderson and Dell'Acqua, 2011).

A phosphorylation site was identified in the first intracellular loop of GluA1 (S567) which is phosphorylated by CaMKII and promotes removal of GluA1 from synapses (Lu et al., 2010, Coultrap et al., 2014). CaMKII is well known to be required for LTP, but its activity is also required during LTD where a prolonged “weak” stimulus favors the phosphorylation of GluA1 S567 over S831, leading to removal of synaptic GluA1 (Coultrap et al., 2014). GluA1 undergoes dephosphorylation of T840 and S845 during LTD by protein phosphatases PP1/PP2A and CaN, respectively (Delgado et al., 2007, Toda and Huganir, 2015, Lee et al., 1998, Sanderson et al., 2016). Under basal conditions, approximately 50% of GluA1 is phosphorylated at T840 and acute inhibition of PP1/PP2A results in very rapid increases in T840 phosphorylation suggesting that this site undergoes rapid and regulated cycling between phosphorylation states (Babiec et al., 2016, Lee et al., 2007). GluA1 T840 phosphorylation has been shown to increase GluA1 single channel conductance while phosphorylation of S845 increases channel open probability, suggesting that dephosphorylation of these sites might also be important to reduce ion channel properties for expression of LTD (Banke et al., 2000, Jenkins et al., 2014).

To summarize, LTD requires a net removal of AMPARs from the post synaptic membrane. In the context of the AMPAR code, we propose that at synapses that contain mixed AMPAR subunits and PTMs, a hierarchy exists such that certain AMPAR variants will be disproportionally targeted for removal during LTD, for example, GluA2/3 containing receptors phosphorylated at S880 (Figure 2). Hippocampal LTD is still observed in knock-outs of GluA2/3 (Meng et al., 2003), but not in neurons expressing GluA2 fused to the GluA1 C-tail (Zhou et al., 2017). Again, these findings suggest that synaptic plasticity is modulated by regulatory elements with the AMPAR C-tails, but this regulation is circumvented in the absence of AMPAR subunits.

AMPAR code and homeostatic scaling-up

GluA1—In response to chronic hyperactivity or hypoactivity, neurons can engage homeostatic scaling-down or scaling-up respectively, in which the strength of excitatory synapses on a neuron is globally modified to maintain neuronal activity in a physiological range (O'Brien et al., 1998, Turrigiano et al., 1998, Turrigiano, 2008). Similar to LTP and LTD, synaptic strength is altered during homeostatic scaling by changing the abundance of

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post-synaptic AMPARs (O'Brien et al., 1998). Homeostatic scaling involves changes in AMPAR protein expression, subunit composition, protein interactions, and PTMs. TTX-induced homeostatic scaling-up in cultured neurons increases GluA1 protein expression as well as surface and synaptic levels of CP-AMPARs (GluA1 homomers) and GluA1/2 heteromers but not in GluA3-containing AMPARs (Figure 3) (Diering et al., 2014, Soares et al., 2013, Tan et al., 2015, Kim and Ziff, 2014, Hu et al., 2010). Scaling-up involves an increase in phosphorylation of GluA1 S845, but not S831, and is completely blocked in neurons with GluA1 S845A knock-in mutations, but remains intact in GluA1 S831A mutant neurons (Diering et al., 2014, Kim and Ziff, 2014). Interestingly, during scaling-up, the overall PKA activity in neurons is reduced. GluA1 S845 phosphorylation is increased by a coordinated recruitment of the remaining active PKA into synapses by the signaling scaffold protein AKAP5, and the simultaneous suppression of CaN activity (Figure 3) (Diering et al., 2014, Kim and Ziff, 2014). In the GluA1 S845A knock-in neurons, TTX still induces an increased expression of GluA1 protein, but this is insufficient to drive scaling-up without the phosphorylation-dependent recruitment of GluA1 into synapses (Diering et al., 2014). Homeostatic scaling-up has also been observed in the visual cortex in mice in response to visual deprivation, and involves the recruitment of synaptic CP-AMPARs and increased GluA1 S845 phosphorylation (Goel et al., 2006, Goel et al., 2011). Like in cultured neurons, visual deprivation-induced scaling-up is absent in GluA1 S845A knock-in mutant mice (Goel et al., 2011).

GluA2—While many studies highlight a role of GluA1 in homeostatic scaling-up, GluA2 and its direct binding partners, GRIP1/2 and PICK1, also play key roles. Scaling-up can be blocked by the expression of the soluble cytosolic C-terminal tail of GluA2 or by shRNA-mediated knock-down of GluA2 (Gainey et al., 2009, Ancona Esselmann et al., 2017). Scaling-up can be rescued by expression of shRNA-resistant WT GluA2, but not by a GluA2 chimera with the C-tail swapped for GluA1, suggesting that elements of the GluA2 C-terminus are required for scaling-up (Gainey et al., 2009). Interestingly, scaling-up can be supported by GluA2 with the C-terminus from GluA1 when S818 is mutated to alanine (GluA2-A1CTD S818A) to mimic the corresponding amino acid in GluA2 (Ancona Esselmann et al., 2017). Scaling-up appears to involve increased tyrosine phosphorylation of GluA2 as a result of downregulation of the tyrosine phosphatase, STEP61 (Figure 3), and scaling-up could be blocked by exogenously applied STEP61 (Jang et al., 2015). Prolonged TTX treatment in cultured neurons causes lysosomal degradation of PICK1, while deletion of PICK1 enhances mEPSC amplitude and occludes homeostatic scaling-up, suggesting that decreased activity of PICK1 is an important mediator of scaling-up (Figure 3) (Anggono et al., 2011). TTX also causes a downregulation of total neuronal GRIP1/2 protein levels but simultaneously causes an increase in the targeting of GRIP1/2 to the PSD which presumably is required for the recruitment of GluA2-containing receptors (Figure 3) (Tan et al., 2015, Gainey et al., 2015). Consistent with these data, deletion of the GRIP1/2 genes completely blocks scaling-up (Tan et al., 2015). Neither GRIP1/2 nor PICK1 are required for scaling-down (Tan et al., 2015, Anggono et al., 2011). Despite the fact that both GluA1 and GluA2 have been implicated in homeostatic scaling-up, complete knock-out of GluA1 still allowed for the scaling-up of GluA2/3, and complete knock-out of GluA2/3 still allowed for the

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scaling-up of GluA1 (Altimimi and Stellwagen, 2013), suggesting that regulatory elements in GluA1/2 can be circumvented by complete removal of these subunits.

AMPAR code and homeostatic scaling-down

GluA2—Bicuculline-induced scaling-down in cultured neurons involves the downregulation of GluA1/2/3 protein levels as well a reduced synaptic targeting of all AMPAR subunits (Diering et al., 2014, Tan et al., 2015). AMPAR association with the PSD is reduced during scaling-down by increasing the rate at which AMPARs dissociate from the PSD scaffold (Tatavarty et al., 2013). This might be partly a result of dephosphorylation of GluA1 S831 and S845, and GluA2 S880 and Y876 (Figure 3)(Diering et al., 2014, Siddoway et al., 2013, Jang et al., 2015). Dephosphorylation of GluA2 S880 during scaling-down is mediated by activation of phosphatase PP1, through its regulatory protein inhibitor 2 (I-2) (Siddoway et al., 2013), while dephosphorylation of GluA2 Y876 requires an upregulation of STEP61 (Jang et al., 2015). Dephosphorylation of these two sites enhances the interaction of GluA2 with GRIP1/2 scaffold proteins (Hayashi and Huganir, 2004, Chung et al., 2000). Concomitantly, GRIP1 protein is upregulated during scaling-down, but remains in intracellular compartments where it may act to sequester internalized GluA2-containing receptors away from the synapse (Figure 3) (Tan et al., 2015). Bicuculline treatment stimulates GluA2 ubiquitination on K882 (Widagdo et al., 2015), very close to the Y876 and S880 phosphorylation sites, and AMPAR ubiquitination has been shown to be required for homeostatic scaling-down (Scudder et al., 2014). Whether GluA2 ubiquitination and phosphorylation interact in the context of homeostatic scaling-down has not been determined (Figure 1).

GluA1—Synaptic GluA1 is dephosphorylated at S831 and S845 during scaling-down and also in the forebrain during sleep (Diering et al., 2014, Diering et al., 2017), which together with other observations suggests that homeostatic scaling-down occurs across the forebrain during sleep (Diering et al., 2017). Decreased phosphorylation of S845 during scaling-down is not mediated by phosphatase activation, but rather by a decrease in PKA targeting to the synapse through uncoupling with the AKAP5 scaffold, favoring the dephosphorylated state of GluA1 (Figure 3) (Diering et al., 2014). This mechanism of decreasing the phosphorylation of GluA1 S845 is completely distinct from the dephosphorylation mediated by acute activation of CaN during LTD (Sanderson et al., 2016). Interestingly, GluA1 which remains phosphorylated is resistant to scaling-down and the unphosphorylated GluA1 containing receptors are preferentially removed (Figure 4) (Diering et al., 2014). Bicuculline treatment drives GluA1 ubiquitination of K868 and is likely part of the mechanism leading to downregulation of AMPAR protein levels during scaling-down (Widagdo et al., 2015, Scudder et al., 2014). It has been shown that phosphorylated GluA1 S845 is resistant to ubiquitination (Guntupalli et al., 2017), offering a potential mechanism for the differential removal of phosphorylated and unphosphorylated GluA1 during scaling-down.

One of the most pertinent questions remaining in the study of homeostatic scaling, is to understand how this form of plasticity interacts with Hebbian plasticity. In order for scaling to proceed in restoring or rebalancing neuronal excitability without catastrophic disruption of memory storage, scaling should be highly sensitive to previous episodes of LTP and LTD.

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As both homeostatic and Hebbian forms of synaptic plasticity involve the trafficking and modification of AMPARs it is completely impossible for the different plasticity types to occur independently. We hypothesize that local forms of Hebbian plasticity may involve some sustained signaling and modification of AMPARs which could influence the expression of subsequent global plasticity. Thus, homeostatic scaling may proceed in a manner which effects the entire neuron while at the same time is locally sensitive to the previous history of individual synapses, allowing neurons to alter their excitability while maintaining information storage. For example, homeostatic scaling-down during sleep should weaken the majority of synapses while retaining the synapses which have undergone LTP while the animal was awake (Diering et al., 2017). Indeed, 3D electron microscopy reconstruction of dendritic spines in mouse cortex comparing wake and sleep showed that ~80% of spines shrink during sleep in a multiplicative manner as predicted by homeostatic scaling-down, whereas the remaining 20% of dendritic spines appear to escape homeostatic scaling-down altogether (de Vivo et al., 2017). The studies on the role of AMPAR phosphorylation during homeostatic scaling suggest that removal of AMPARs from synapses during scaling-down may follow a hierarchy whereby dephosphorylated receptors are preferentially removed. Synapses that have undergone LTP and maintain some signaling activity and populations of phosphorylated AMPARs may be protected from scaling down (Figure 4). Further exploration of the concepts of the AMPAR code will be important to understand the interaction of Hebbian and homeostatic plasticity as they push and pull on the very same post-synaptic AMPARs.

AMPAR code and the post-synaptic density slot hypothesis

A prominent model of synaptic strength posits that the post-synaptic density (PSD), a dense network of synaptic scaffolding, adhesion and cytoskeletal proteins, contains ‘slots’ that might be occupied by AMPARs supplied by lateral diffusion from the peri-synaptic region (Opazo et al., 2012, Huganir and Nicoll, 2013, Kessels and Malinow, 2009). Surface localized AMPARs are supplied by exocytosis from pools of intracellular receptors contained within endosomes (Lin et al., 2009, Makino and Malinow, 2009). The concentration of surface, extra-synaptic AMPARs is controlled globally and locally by constitutive or activity-dependent exocytosis/endocytosis respectively. In the plane of the plasma membrane, AMPARs are highly mobile but become trapped in PSD slots (Opazo et al., 2012). AMPARs eventually diffuse out of the PSD slots and can then be removed from the cell-surface via endocytosis. During synaptic plasticity, the number of PSD slots may be altered, and/or the affinity of PSD slots for AMPARs might be changed, such that AMPARs can accumulate in the PSD during LTP and are removed during LTD (Figure 2). The molecular basis of the slot is typically attributed to intracellular PSD scaffold proteins binding to PDZ ligands on AMPARs or auxiliary subunits (Hayashi et al., 2000, Sheng et al., 2018), but may also involve transmembrane, trans-synaptic, or secreted proteins interacting with the extracellular AMPAR amino terminal domains, including the secreted neuronal pentraxins (Watson et al., 2017, Diaz-Alonso et al., 2017, Sia et al., 2007, O'Brien et al., 1999, Lee et al., 2017) (Fig. 2). Under basal conditions the number and structure of the slots may be highly stable, while in the minutes following LTP/LTD induction, the PSD slots as a whole may be transiently malleable, not simply added or subtracted, such that local AMPAR dynamics become critical.

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The AMPAR code predicts that under certain conditions the PSD slots will have biased affinity towards certain AMPAR variants, for example phosphorylated GluA1-containing receptors during noradrenaline facilitated hippocampal LTP (Makino et al., 2011, Hu et al., 2007). An alternative, which is not mutually exclusive, is that occupation of the PSD slots will be heavily influenced by the AMPARs which are most readily available on the cell surface, likewise favoring certain AMPAR variants (Figure 2). Large packets of GluA1 are inserted into the dendritic shaft from intracellular recycling endosomes, containing an estimated fifty GluA1-containing receptors, in an activity-dependent manner (Lin et al., 2009, Makino and Malinow, 2009). The majority of these GluA1-containing receptors are likely to be GluA1/2 heteromers (Wenthold et al., 1996, Lu et al., 2009). Activity-dependent increases in extra-synaptic GluA1 have also been observed *in vivo* in the mouse barrel cortex in response to whisker stimulation (Zhang et al., 2015). GluA2-containing receptors (likely GluA2/3 heteromers) undergo spontaneous and constitutive exocytosis, but these events are much smaller, containing an estimated 2-5 receptors (Araki et al., 2010, Gu et al., 2016, Lin et al., 2009). As new PSD slots are formed during LTP, or as existing slots increase their affinity for AMPARs, the large and localized activity-dependent increase in extra-synaptic GluA1 favors the accumulation of GluA1 at potentiated synapses. During LTD, AMPARs must be removed from the PSD, and this step seems to be tightly coupled to removal of the AMPARs from the plasma membrane through clathrin-dependent endocytosis. Block of endocytosis has been shown in many studies to prevent LTD, suggesting that AMPARs must be locally depleted from the cell surface, perhaps to prevent rapid refilling of the PSD-slots. GluA2 seems to play an especially important role in LTD (Zhou et al., 2017). GluA2 can directly interact with AP-2 clathrin adapter complex, which allows GluA2-containing receptors to be rapidly depleted from the cell-surface (Kastning et al., 2007, Lee et al., 2002). Thus the “slot hypothesis” offers some explanation for the prominent role of GluA1 in LTP and of GluA2 in LTD, via the rapid activity-dependent mobilization of GluA1-containing receptors onto the cell surface during LTP, and the removal of GluA2 containing receptors from the cell-surface during LTD through interactions with AP2 and/or through interactions involving the GluA2 Y-rich motif (Ahmadian et al., 2004, Anggono and Huganir, 2012, Huganir and Nicoll, 2013).

Forms of LTP, LTD and homeostatic scaling have been shown to be intact in knock-out mice of different AMPAR subunits as well as in using molecular replacement experiments (Huganir and Nicoll, 2013, Granger et al., 2013, Ancona Esselmann et al., 2017). Where does this leave the AMPAR code of synaptic plasticity? How, for example, is LTD and homeostatic scaling-up blocked in GluA1 S845A knock-in mice, while both forms of plasticity are intact in the complete absence of GluA1 (Lee et al., 2010, Diering et al., 2014, Altimimi and Stellwagen, 2013, Selcher et al., 2012)? We propose that through subunit specific PTMs and protein interactions, layers of regulation are built into the AMPARs such that each form of synaptic plasticity will result in the recruitment or removal of the correct types and number of AMPARs from PSD slots (Zhou et al., 2017). The recently demonstrated subunit-specific roles of AMPAR amino-terminal domains may also inform the molecular basis of these synapse slots (Watson et al., 2017, Diaz-Alonso et al., 2017, Sia et al., 2007). This recruitment or removal follows a hierarchy, with biased trafficking, accumulation or removal of certain AMPAR variants (Figure 2). In the case where individual

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AMPA subunits are absent, the layers of regulation associated with that subunit are also removed, and previously unfavorable AMPARs in the hierarchy for that plasticity type are "promoted" in the ranking allowing for synaptic plasticity to proceed (Granger et al., 2013) (Figure 2).

Concluding remarks

AMPARs play an essential role in the functioning of the brain, and changes in the abundance of post-synaptic AMPARs are a fundamental mechanism for most forms of synaptic plasticity (Huganir and Nicoll, 2013). Previously, it was proposed that combinatorial modifications of histones could regulate chromatin structure and gene expression, considerably expanding the informational content of DNA (Jenuwein and Allis, 2001). While the translation of this "code" is far from complete, the concept has proven to be very influential in the field of epigenetics (Taverna et al., 2007). The AMPAR code likewise suggests that the informational content of synapses will include the combinatorial effects of AMPA subunits, PTMs and binding partners. Knock-out or molecular replacement studies have championed the slot hypothesis (Granger et al., 2013). Such studies have revealed the remarkable ability of synaptic plasticity to resist manipulation. However, studies using more subtle approaches in which individual sites of AMPAR modification are altered support the AMPAR code hypothesis by showing that plasticity and animal behavior are usually sensitive to sophisticated regulatory mechanisms acting on AMPARs, for example, phosphorylation of GluA1 during noradrenaline mediated hippocampal LTP (Makino et al., 2011, Hu et al., 2007, Zhou et al., 2017). Here, we have only begun the translation of the AMPAR code. The 11 known PTMs of GluA1 alone provides the possibility of 2,048 possible variants, each of which may alter synaptic function or plasticity in subtle ways. Furthermore, synaptic plasticity is often studied one form at a time, one synapse at a time. The emerging prominence of circuit dynamics in information processing in the brain shows us that complex animal behaviors involve synaptic plasticity throughout neuronal circuits, simultaneously involving many synapses. Complex processes such as memory consolidation are also likely to involve multiple forms of synaptic plasticity including both Hebbian and homeostatic plasticity (Diering et al., 2017). Continued efforts in translation of the AMPAR code will be essential to understand how these multiple forms of synaptic plasticity not only coexist but are well coordinated to allow for the emergence of higher cognition.

Acknowledgements

We would like to thank Dr. Natasha Hussain and Dr. Hana Goldschmidt for critical reading of this manuscript and for insightful discussion.

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Changes in the number and properties of postsynaptic AMPARs is a fundamental mechanism of synaptic plasticity. Diering and Huganir discuss how AMPAR subunits, posttranslational modifications, and protein binding partners form combinatorial molecular codes relevant for understanding synapse function and plasticity.

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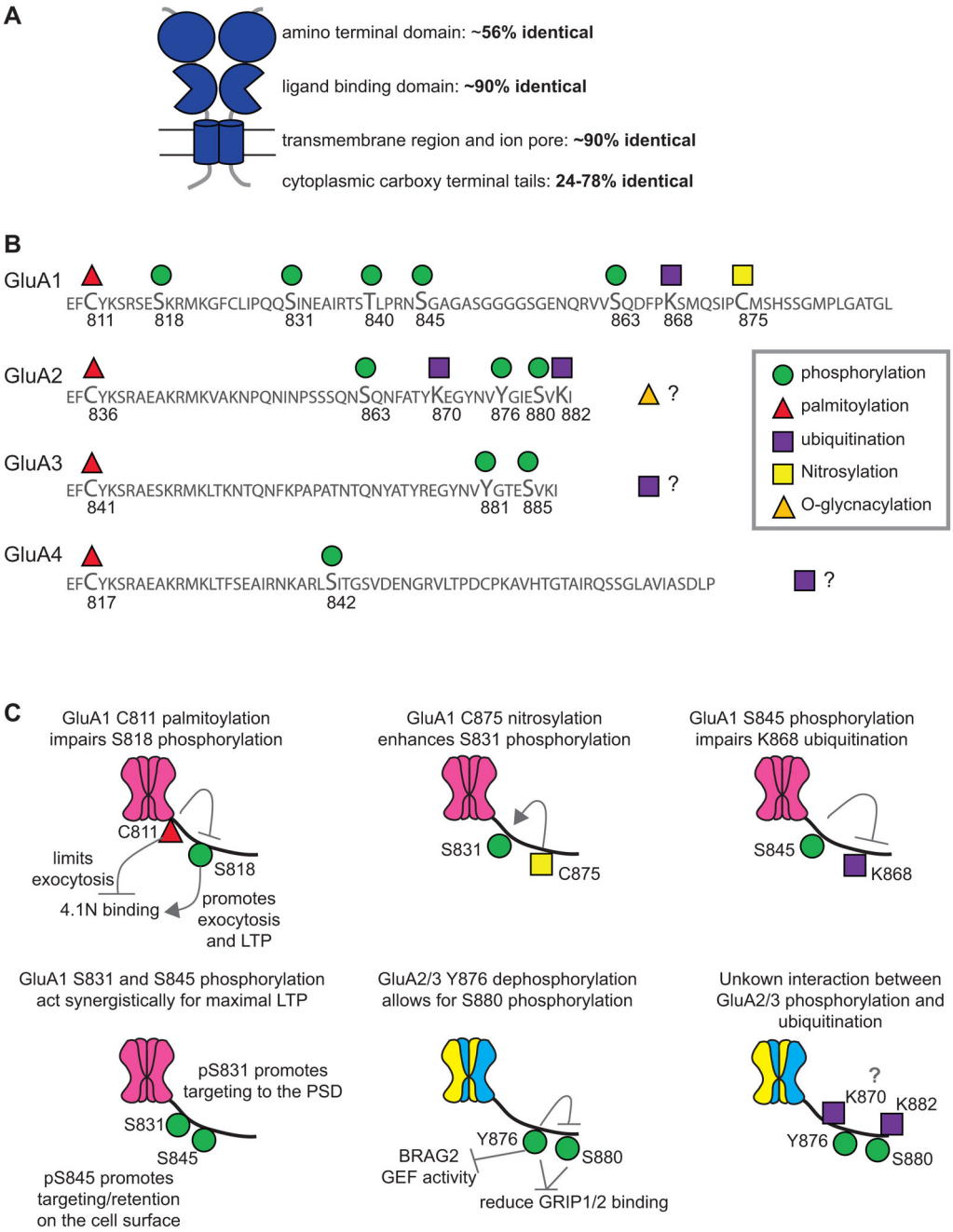

Figure 1.

AMPAR post-translational modifications and the AMPAR code. **A**. Schematic of AMPAR structure. % identity indicates the identity of each domain between the GluA1-4 subunits. **B**. Amino acid sequences of GluA1-4 C-terminal tails with known post-translational modifications indicated. Modifications with a "?" are known to occur but the modified amino acid has not been identified. **C**. Schematic of known interactions between AMPAR post-translational modifications.

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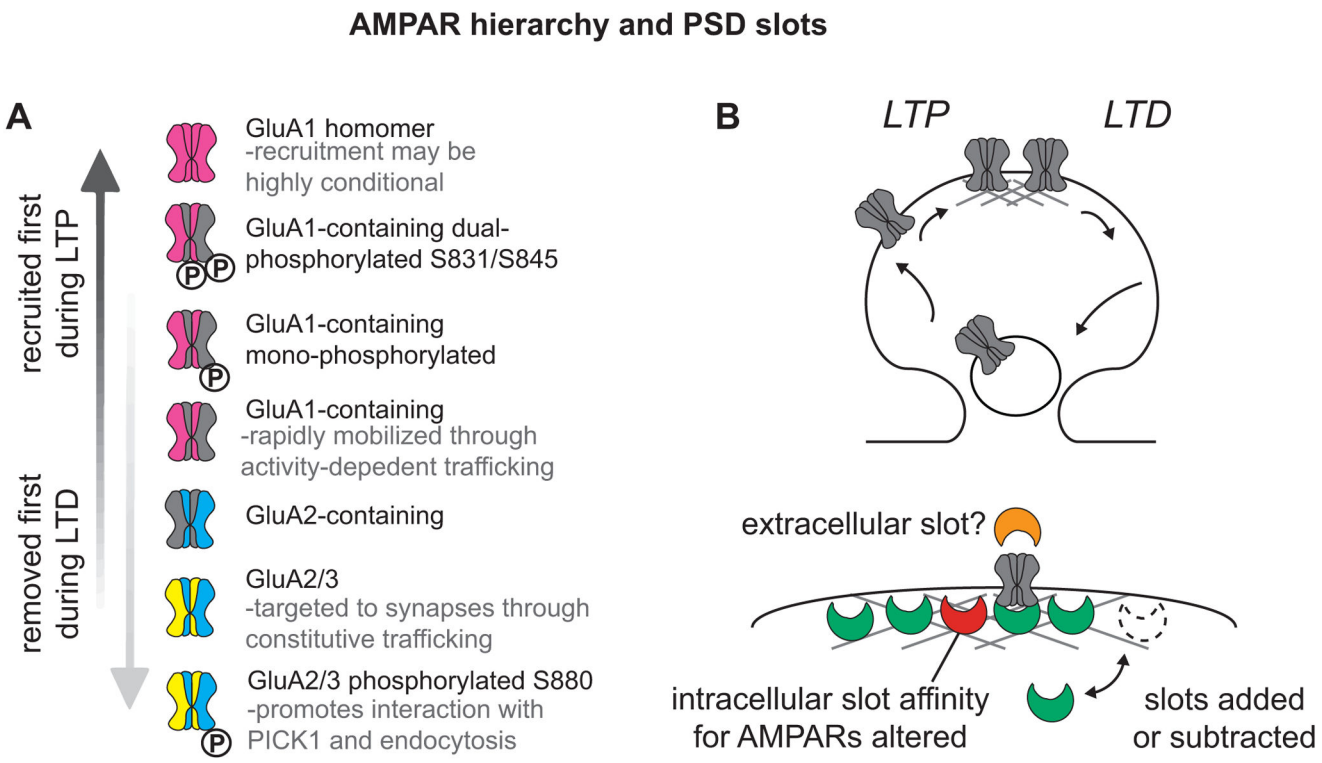

Figure 2.

The AMPAR hierarchy and PSD slot hypothesis. During synaptic plasticity AMPARs are added or subtracted from PSD/extracellular slots (**A**), which are modified in number and/or in their affinity for AMPARs during plasticity (**B**). Trafficking AMPARs then populate these PSD slots in a hierarchy based on their i) affinity for the PSD slots, and ii) their mobilization and mobility on the cell-surface. AMPARs enter the PSD slots through lateral diffusion from extra-synaptic pools of surface receptors. During LTP, surface pools are locally enriched with GluA1-containing AMPARs through activity-dependent exocytosis, whereas during LTD surface AMPARs are locally depleted through endocytosis of GluA2-containing receptors.

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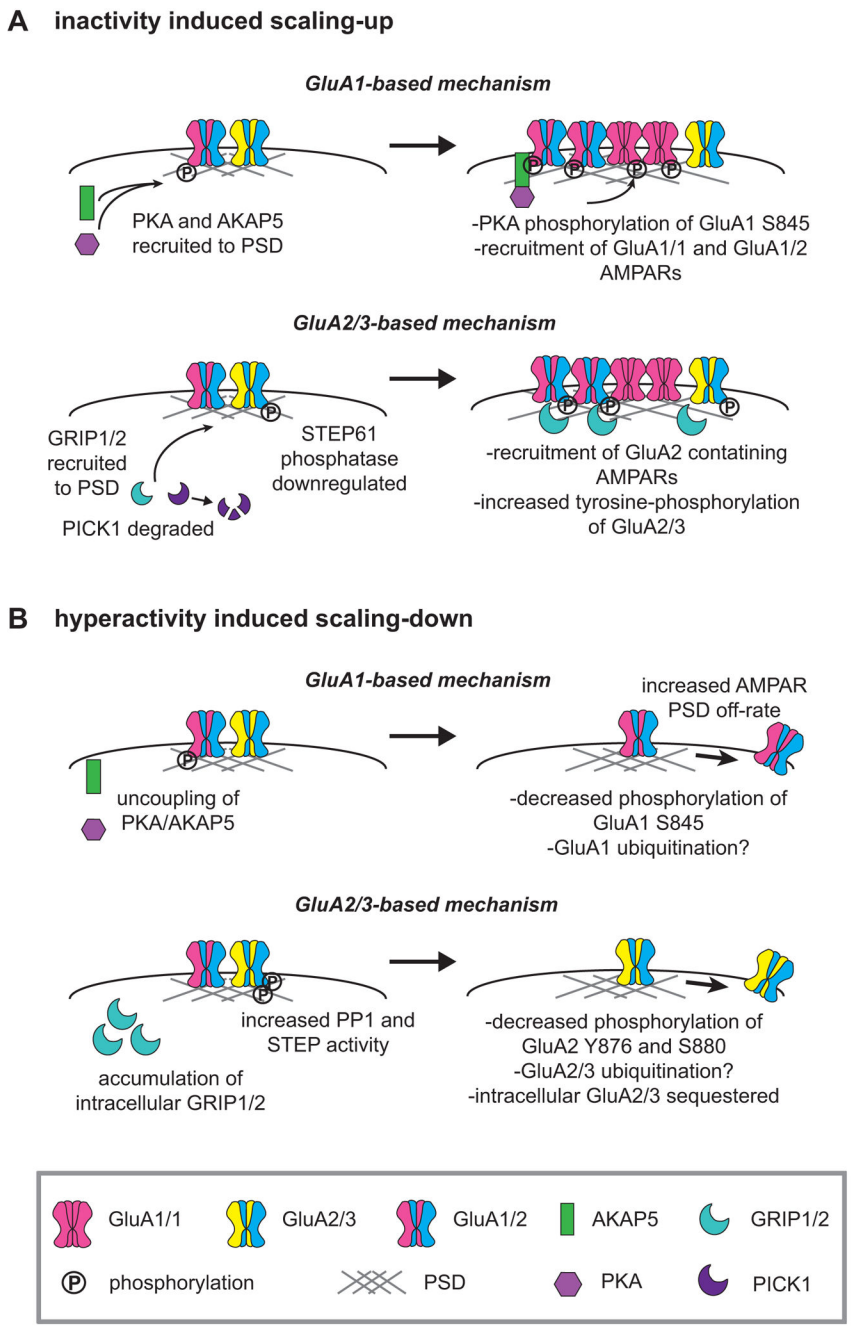

Figure 3.

Subunit-specific mechanisms of homeostatic scaling. **A and B**, schematics of known mechanisms of homeostatic scaling-up (**A**) and scaling-down (**B**) based on GluA1 or GluA2/3.

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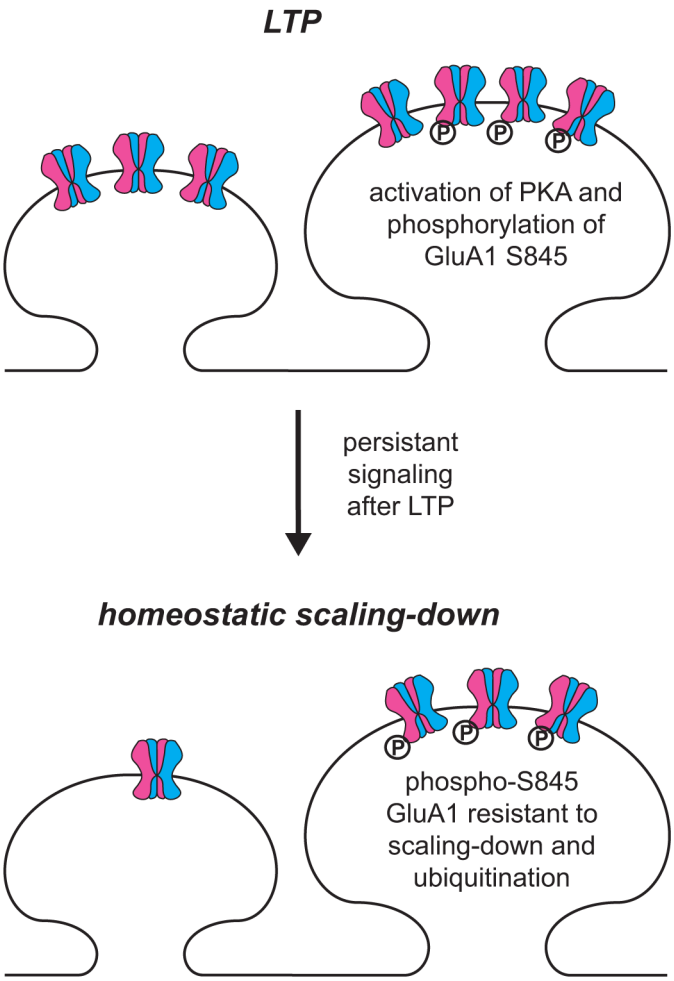

Figure 4.

Communication between Hebbian and homeostatic plasticity based on the AMPAR code. AMPARs are modified at synapses which have undergone Hebbian plasticity, for example phosphorylation of GluA1 during LTP. These synapses maintain signaling which sustains a population of modified receptors. During homeostatic scaling, synapses across the neuron are strengthened or weakened by adding or subtracting AMPARs respectively. This global plasticity is modified locally by the population of modified AMPARs which are either resistant or hypersensitive to homeostatic scaling. This interaction allows global homeostatic plasticity to be sensitive to previous history of Hebbian plasticity.